

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			$[H^+] = 10^{-3.24}$ (1) $[H^+] = 5.75 \times 10^{-4} \text{ mol dm}^{-3}$ (1)		2		2	2	
	(b)	(i)		filtration		1		1		1
		(ii)	I	excess sodium hydroxide dissolves any precipitate produced by zinc (1) excess ensures that all iron ions are precipitated (1)			2	2		2
			II	moles $\text{Fe}_2\text{O}_3 = \frac{14.5 \times 10^{-3}}{159.6} = 9.085 \times 10^{-5} \text{ mol}$ (1) concentration = $\frac{2 \times (9.085 \times 10^{-5})}{0.025} = 7.27 \times 10^{-3} \text{ mol dm}^{-3}$ (1)		2		2	2	
			III	$\frac{0.2}{14.5} \times 100 = 1.4\%$		1		1	1	
		(iii)	I	solution turns (pale) pink do not accept purple to pink	1			1		1
			II	moles of $\text{MnO}_4^- = 3.80 \times 10^{-5} \text{ mol}$ (1) $[\text{Fe}^{2+}] = 7.60 \times 10^{-3} \text{ mol dm}^{-3}$ (1) ecf possible		2		2	2	

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		(iv)		award (1) for percentage difference between the two values $\Rightarrow$ either 4.3% or 4.5% (1) students' decision is not valid award (1) for sensible explanation e.g. - total of percentage errors for both methods is 2.2% (1.4% + 0.8%) and the percentage difference is greater than this - total of percentage errors in titration and gravimetric analysis allow a difference of only 0.16×10^{-3} - values are $7.60 \pm 0.10 \times 10^{-3}$ and $7.27 \pm 0.06 \times 10^{-3}$ and these ranges do not overlap			2	2	2	
	(c)	(i)		royal blue / dark blue do not accept blue / pale blue or any reference to a precipitate	1			1		1
		(ii)		Indicative content - A negative ΔG means products are more likely to be present than reactants - Higher product concentration leads to large value of K_c - Formation of the Cu-cyclen complex breaks four Cu-ammonia bonds and forms four Cu-cyclen bonds - As the bonds broken and bonds made are of similar strengths enthalpy change will be small - For a negative ΔG, ΔS must be positive OR $T\Delta S$ must be greater than ΔH as $\Delta G = \Delta H - T\Delta S$ - The forward reaction produces more (aqueous) molecules as four NH_3 are released - The entropy of free ammonia would be higher than cyclen so ΔS would be positive		2	4	6	1	

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				5-6 marks The candidate includes five points with valid discussion of entropy, enthalpy and the position of equilibrium <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The information included in the response is relevant to the argument.</i> 3-4 marks The candidate includes four points with valid discussion of two of entropy, enthalpy and the position of equilibrium <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. Mainly relevant information is included in the response but there may be some minor errors or the inclusion of some information not relevant to the argument.</i> 1-2 marks The candidate includes two relevant points <i>There is a basic line of reasoning which is not coherent, supported by limited evidence and with very little structure. There may be significant errors or the inclusion of information not relevant to the argument.</i> 0 marks No attempt made or no response worthy of credit.						
				Question 10 total	2	10	8	20	10	5